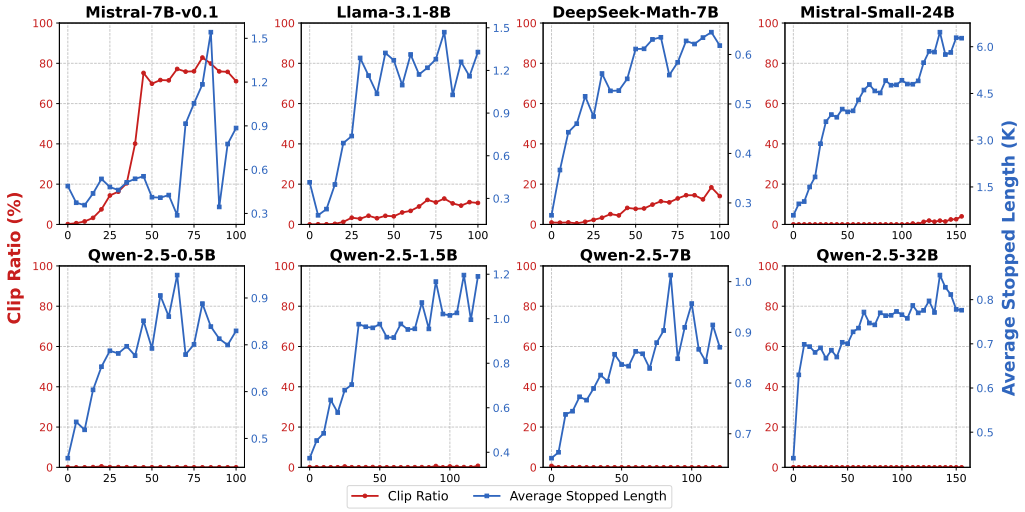


Figure 4: Average clip ratio and stopped length across training iterations for different models. We assess the models every five steps on a variety of math benchmarks, including GSM8K, MATH500, Minerva Math, and OlympiadBench, as well as competition-level benchmarks like AIME24 and AMC23. The red line indicates the clip ratio, while the blue line represents the stopped length. Per-benchmark results are in Figure 12 (Appendix D).

Smaller models, such as Qwen-2.5-0.5B and 1.5B, tend to prioritize learning the "Subgoal Setting" behavior, with its proportion increasing by approximately 4–5 times. Additionally, the proportions of "Verification" and "Enumeration" also show noticeable growth. In contrast, for other base models that inherently possess step-by-step reasoning capabilities, adjustments in "Subgoal Setting" during the RL training process are relatively minor.

DeepSeek-Math-7B, Llama-3.1-8B, and Mistral-Small-24B exhibit substantial increases in the proportions of "Enumeration" and "Verification" behaviors, rising from relatively low initial levels by approximately 3-4 times. This growth correlates closely with their changes in average stopped length, suggesting a shift in reasoning patterns over time. For instance, in Mistral-Small-24B, reflection-oriented behaviors such as "Verification" and "Backtracking" increase dramatically from nearly 0% to approximately 50%, indicating the emergence of reflection behavior from scratch. This shift suggests that the model progressively internalizes verification as part of its reasoning process, offering a promising trajectory for enhancement.

In contrast, Qwen-2.5-7B and 32B demonstrate strong reasoning behaviors from the outset, with minimal changes throughout training. This phenomenon aligns with their slow length adjustments (Figure 1) and suggests that Qwen models inherently possess robust reasoning capabilities. Rather than undergoing a structural shift in their reasoning processes, they primarily benefit from small increases in thinking time, which yield significant performance improvements. Finally, we observe that Mistral-7B-v0.1 consistently exhibits low reasoning behaviors with no noticeable growth, further supporting our earlier analysis in §2.3.

To intuitively illustrate the changes in reasoning behavior, we present examples of Mistral 24B's reasoning before and after training in Figures 22. Comprehensive case studies involving other models are available in Appendix I.3. In Figure 22, we observe that unlike the base model, the zero training model actively attempts to verify if its initial solution is valid by substituting it back into the original expression. Upon recognizing that the first solution does not meet the necessary conditions, the model explicitly initiates a backtracking approach, stating "let's try another possibility," eventually arriving at the correct answer.

3 Key Factors Shaping Zero RL Training

In this section, we identify key factors that influence stability and performance during zero RL training, particularly when dealing with early-stage or weaker models. First, we explore how an over-reliance on format rewards restricts exploration. Next, we analyze how

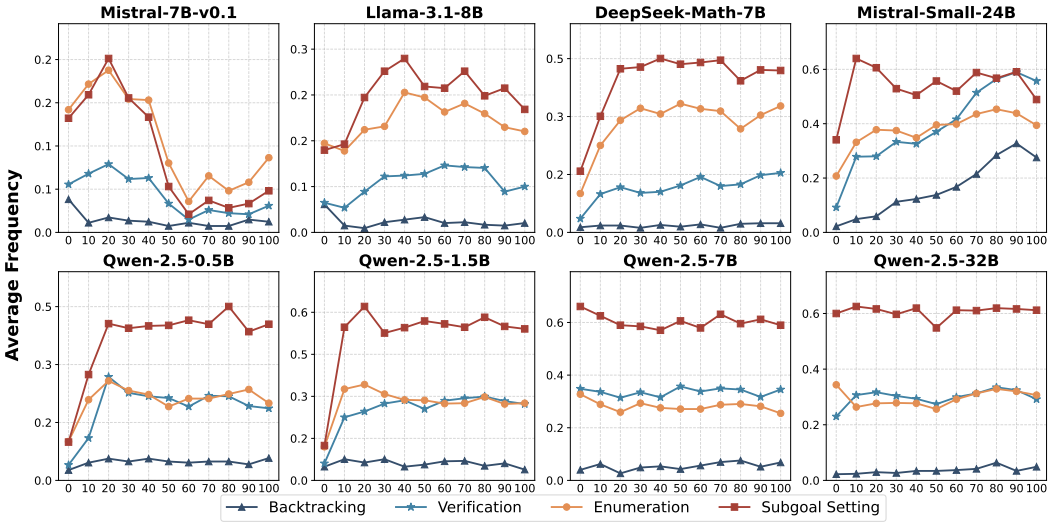


Figure 5: The change in reasoning behavior over the training iterations across all models. As described in §2.2, we use GPT-4o to extract and track shifts in reasoning behaviors on OlympiadBench. We focus on four reasoning-related behaviors: "Backtracking", "Verification", "Subgoal Setting", and "Enumeration".

data difficulty impacts exploratory behavior, illustrating how exposure to varying levels of difficulty shapes the exploration dynamics. We also discuss the impact of exploration-related hyperparameters in Appendix G.

3.1 Over-Reliance on Format Rewards

We find that enforcing strict formatting constraints, such as requiring the final answer to be enclosed in a latex command $\backslash boxed\{\}$, can hinder model's freely exploration and ultimately degrades performance. This is because many base models cannot follow the format constraint well in the initial stage, and imposing a format reward will penalize many correct explorations. We compare two reward functions: one without format constraints, which rewards responses solely based on answer correctness (our default design in §2.1), and another that strictly enforces formatting by penalizing responses with a reward of -1 if they fail to adhere to the required format.

Figure 6 illustrates weaker models like Llama-3.1-8B struggle under strict formatting requirements, leading to a rapid increase in response length early in training without performancec improvement. The model expends excessive effort on adhering to the format but fails to learn how to answer correctly, ultimately resulting in model collapse. Figure 6 (Left) further reveals that even stronger models, such as Qwen-2.5-7B, which initially comply with formatting constraints, suffer in later training stages. This includes both performance degradation and a significant reduction in CoT length. These findings highlight that: in a zero RL training setting, rather than imposing rigid formatting rules, we should prioritize maintaining response verifiability while allowing sufficient flexibility for exploration.

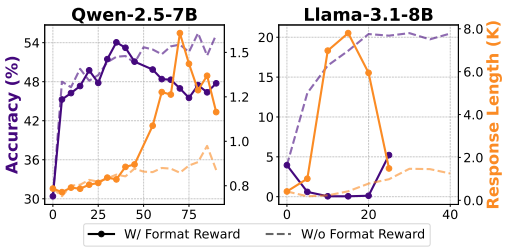
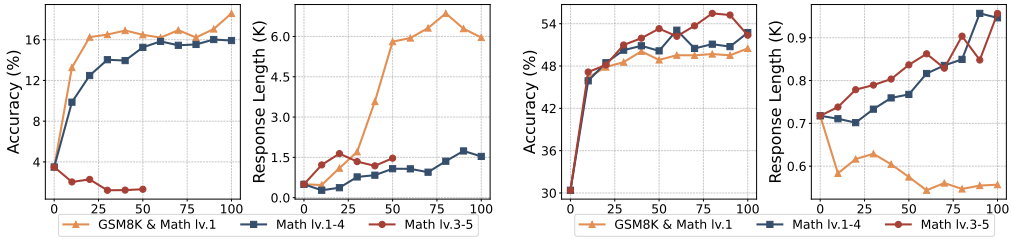


Figure 6: Accuracy and response length with and without format rewards.



(a) Mistral-7b-v0.1 (b) Qwen-2.5-7B

Figure 7: Comparison of accuracy and response length across different data difficulty levels. We examine three levels of data: Easy (GSM8K and MATH lv.1), Medium (MATH lv.1–4), and Hard (MATH lv.3–5), with each category containing approximately 8,000 problems.

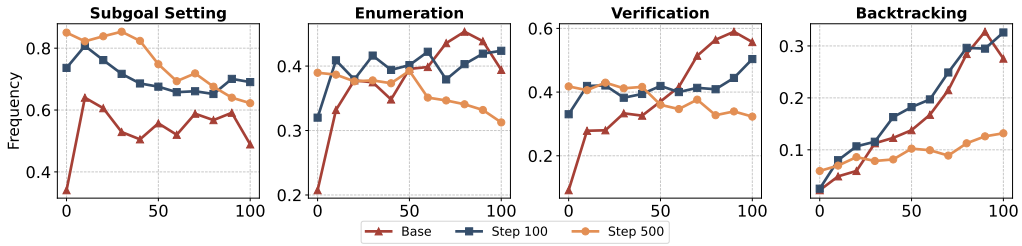


Figure 8: Reasoning behavior ratio over RL training iterations after using different SFT steps as starting points. "Base" refers to the base Mistral-Small-24B model without any SFT, while "Step 100" and "Step 500" represent 100 and 500 steps of SFT, respectively. As described in §2.1, we use GPT-4o to track shifts in reasoning behaviors on OlympiadBench.

3.2 Data Difficulty on Exploratory Behavior

Base models exhibit varying performance and CoT behaviors when trained on different RL data. Figure 7 compare the performance of Mistral-7B and Qwen-2.5-7B across Easy, Medium, and Hard datasets. We observe a clear trend: as data difficulty increases, Mistral-7B's performance progressively deteriorates. When faced with high-difficulty data (Hard: MATH levels 3-5), the model struggles to generate responses that receive positive feedback from the reward system. This failure results in a significant increase in response length without any corresponding improvement in accuracy, signaling a breakdown in the training process—often referred to as training collapse. Figure 7 Left demonstrates that Qwen-2.5-7B exhibits a pattern entirely opposite to Mistral-7B-v0.1. Specifically, as dataset difficulty decreases, both the model's average accuracy and response length decline, with the effect being most pronounced on the simplest dataset, where even response length decreases. This finding aligns with our previous analysis of Qwen-2.5-7B in §2.4, reinforcing the notion that Qwen inherently possesses strong reasoning capabilities. To further improve its response length, training should incorporate more challenging datasets to encourage deeper reasoning and extended thinking time. The analysis highlights that zero RL training data must align with the base model's inherent reasoning capabilities.

4 Revisiting Traditional SFT for RL-Driven Reasoning Emergence

As base models may not follow instruction well and pose a major challenge for zero RL training, one may wonder a simple SFT stage as a cold start may be helpful to learn to follow instructions well. In this section, we revisit the impact of traditional SFT methods (where the responses are not from long CoT models) as a cold start on RL training performance and reasoning behavior–notably, this was the most commonly used post-training pipeline with RL following an SFT stage, before DeepSeek-R1. Specifically, we use a subset of